

Cambridge (CIE) IGCSE Additional Maths Paper 1: Non-Calculator (Set A)

Sunday 5 October 2025

Afternoon (Time: 2 hours 0 minutes)

Total marks

/ 80

Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- Answer all questions.
- You must show all necessary working clearly.
- Calculators must not be used in this paper.

Materials

- [List of formulas](#)

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- 1 (a)** Two functions are given by $f(x) = 1 + 8x$ where $x \in \mathbb{R}$ and $g(x) = 2 + \ln x$ where $x > 0$.

Find an expression for $f^{-1}(x)$.

(1 mark)

- (b)** State the geometric relationship between the graph of $y = f(x)$ and the graph of $y = f^{-1}(x)$.

(1 mark)

- (c)** Find $fg^2(1)$.

(2 marks)

- (d)** Explain why the function $gf(x)$ does not exist.

(1 mark)

2 (a) The unit vectors $\mathbf{i}$ and $\mathbf{j}$ represent due east and due north respectively.

Person A starts at a position of $(-4\mathbf{i} + \mathbf{j})$ metres and walks with a constant velocity of $(3\mathbf{i} - \mathbf{j})$ metres per second.

Find the position vector of person A after 5 seconds.

(1 mark)

(b) Person B walks with a constant velocity of $(2\mathbf{i} + 2\sqrt{3}\mathbf{j})$ metres per second.

Find

(i) the speed of person B,

(ii) the bearing of the direction in which person B is walking.

(5 marks)

3 (a) Find the values of the constant k for which the equation

$$2kx^2 + (k - 6)x - 4 = 0$$

has no real solutions.

(5 marks)

(b) Hence determine the number of points of intersection, if any, between the line $y = 6x$ and the curve $y = 2kx^2 + kx - 4$ for $k > 0$.

(1 mark)

4 (a) Solve the following equation.

$$\frac{1 + 3e^x}{e^x + e^{-x}} = 1$$

Give your answer(s) in the form $p \ln q$ where p and q are integers.

(4 marks)

(b) Given that $\log_2 y = c$, express $\log_2(2y) - \log_y 2$ in terms of c only.

(2 marks)

- 5 (a)** The graph $y = p(x)$ where $p(x) = 5x^3 - 17x^2 + 16x + a$ intersects the x -axis at the points $(b, 0)$, $(1, 0)$ and $(c, 0)$ where $b < c$.

Find the constants a , b and c .

(5 marks)

- (b)** The curve $y = p(x)$ has a local maximum point at $\left(\frac{2}{3}, \frac{16}{27}\right)$ and a local minimum point at $\left(\frac{8}{5}, -\frac{36}{25}\right)$.

Find the value of d for which the equation $|p(x)| = d$ has exactly 5 distinct solutions.

(2 marks)

- 6** A sphere of radius r cm and volume V cm³ is increasing in size with time t seconds. The volume increases at a constant rate of 24 cm³ s⁻¹.

Find the exact rate at which the radius is increasing when the sphere reaches a volume of $\frac{32\pi}{3}$ cm³.

(4 marks)

7 (a) In this question, all angles are measured in radians.

The graph of $y = a \sin bx + c$ has an amplitude of 10, a period of 16π and passes through the point with coordinates $(12\pi, 4)$.

Find the constants a , b and c .

(4 marks)

(b) Sketch the graph of $y = \left| \tan\left(\frac{x}{2}\right) + 1 \right|$ for $-\pi < x < \pi$.

Label clearly any points of intersection with the coordinate axes and any asymptotes.

(4 marks)

8 (a) A curve has the equation $y = e^{(4x - x^2)}$.

Find and factorise an expression for $\frac{dy}{dx}$.

(2 marks)

(b) Hence find $\int \frac{(2 - x)e^{4x}}{e^{x^2}} dx$.

(2 marks)

(c) Show that the second derivative of the curve $y = e^{(4x - x^2)}$ satisfies the relationship

$$\frac{d^2y}{dx^2} = (p + qx + rx^2)y$$

where p , q and r are constants to be found.

(4 marks)

- 9 (a)** The circle C_1 has a centre at $(3, 4)$ and a radius of 3.

Write down the equation of C_1 .

(1 mark)

- (b)** The circle C_2 has the equation

$$x^2 + y^2 - 12x - 16y + 100 - k^2 = 0$$

where $k > 0$.

Given that C_2 is tangent to C_1 , find the value of k in each of the following cases:

- (i) The circles touch each other once and neither circle lies inside the other.
- (ii) The circles touch each other once and C_1 lies inside C_2 .

(7 marks)

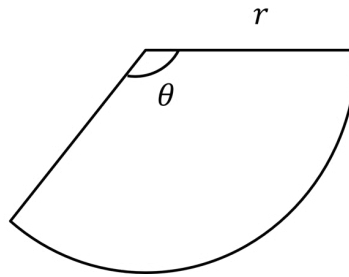
- (c)** A different circle, C_3 , intersects C_1 at two distinct points, one of which has the coordinates $(w, 7)$.

The equation of the common chord to C_1 and C_3 is $x + y = 10$.

Find w .

(1 mark)

10 (a) A sector from a circle of radius r has an internal angle of θ radians, as shown below.



The perimeter of the sector is 4 units and the area of the sector is A square units.

Show that $A = \frac{8\theta}{(2 + \theta)^2}$.

(3 marks)

(b) Show that $\frac{dA}{d\theta} = \frac{p(q - \theta)}{(2 + \theta)^3}$ where p and q are constants to be found.

(3 marks)

(c) Find the maximum area of the sector. You must show that this area is a maximum.

(4 marks)

11 (a) Throughout this question, $A = \frac{\tan x}{1 + \sec x}$ and $B = \frac{\tan x}{1 - \sec x}$.

Show that $AB = k$ where k is a constant to be found.

(2 marks)

(b) Show that $A + B = -2\cot x$.

(4 marks)

(c) Use your answers to part (a) and part (b) to simplify the expression $(A + 1)(B + 1)$.

Hence, or otherwise, solve

$$\left(\frac{\tan(x - 60)}{1 + \sec(x - 60)} + 1 \right) \left(\frac{\tan(x - 60)}{1 - \sec(x - 60)} + 1 \right) = 2\sqrt{3}$$

for $-180^\circ < x < 180^\circ$.

(5 marks)